ERC Starting Grant 2021

Research proposal [Part B1]

Sleep and Inertial-based Biomarkers for Alzheimer's Detection in Mild Cognitive Impairment (MCI) Scenarios

SInBAD

Cover page:

- Name of the Principal Investigator (PI): Alberto Corbi Bellot
- Name of the PI's host institution for the project: Universidad Internacional de La Rioja (UNIR)
- Proposal duration in months: 60

Alzheimer's disease (AD) is a type of mental decay affecting over 50M of people in developed countries. Symptoms start with forgivable attention/memory lapses (*amnestic mild cognitive impairment* or MCI) and may later evolve towards dementia. The precise cause of this decline is still unknown, and although a decent level of understanding has been achieved to date, science has not been able to bring definitive treatments nor provide biomarkers of disease progression.

Recent discoveries have intimately linked Alzheimer and sleep. Briefly, during deep sleep the lymphatic system acts as a cleansing mechanism that sieves the poisonous conglomerates that are connected to AD. From a clinical perspective, **the evolution of AD is also appreciable in sleep patterns**. This has been **confirmed** by caregivers and quantitatively measured in controlled experimental settings. However, in spite of these sharp AD⇆sleep prodromal evidence, no solid diagnostic markers (derived from nightly behaviours) have been yet devised. As an active member of the scientific and AD-awareness communities, this **lack of resources in the translational continuum** has called my attention.

The goal of the SInBAD project is to refine a **translational biomarker framework for AD grounded on modern science of sleep**. These time markers (**chronofiducials**) will in turn be based on the **objective**, **unobtrusive**, continuous and **respectful** measurement of nightly behaviours, sleep disruption and **inertial motion patterns** of volunteers who have been formerly diagnosed with MCI and may progress towards full AD. Initially, modern long-term bedtime tracking equipment will be used. Next, **state-of-the-art machine learning methods** will be exploited, with an emphasis on those associated to time series analysis, **causal inference**, weather **forecasting** and climate **prediction**. SInBAD is a multidisciplinary proposal and will be carried out in **collaboration with mental health institutions**, **sleep scientists** and **artificial intelligence** experts.

Explain and justify the cross-panel or cross domain nature of your proposal, if a secondary panel is indicated in the online proposal submission forms. There is a limit of 1000 characters, spaces and line breaks included.

Section a: Extended Synopsis of the scientific proposal

Mild cognitive impairment: time to act while you sleep

Alzheimer's disease (AD) currently affects ~50 million people in developed countries [1]. Despite decades of (fruitful) research, we still do not have a treatment/cure for it. Although there are several theories about its origin, AD is known to be closely related to the massive **abnormal accumulation of a substance called amyloid-β** (Aβ) in the brain. Structures with this product are naturally produced by the body. If they are not properly cleaned away, they end up forming sticky plaques that eclipse synapses. Over time, this accumulation may reach a tipping point in which cellular damage takes place with functional consequences. To make things worse, a crucial neurotransport protein called τP enters to scene becoming hyper-phosphorylated and forming tangles that further contribute to the biological destruction of neural networks.

Given the current lack of treatment for AD, the best strategy to counteract the condition is **prevention**. Sadly, it is very difficult to identify who is going to eventually develop this disease. Experience tells us that it will very likely start with a process known as **amnestic mild cognitive impairment** or **MCI**. MCI is the stage between the expected cognitive decline of normal ageing and a more serious mental deterioration (like that caused by AD). It is mainly characterised by casual short-term memory loss episodes that may increase in relevance/frequency. Within this phase, a person is still at nearly the max brainpower (except for annoying bouts of forgetfulness). Nevertheless, **MCI** can also be understood as a **clinically profitable stage** in which it may be possible to **foresee the eventual advent of AD**. In this period, science is still on time to trigger the implementation of available therapies that **foster the cognitive reserve**. In other words, MCI represents a right time to act and as an **opportunity** not to be missed to detect AD.

In this context, the goal of the SInBAD project is to define a new set of **biomarkers for early Alzheimer's disease (AD) diagnosis that would allow preventive interventions**. This fiduciary system is in turn based on the raw-measured inertial information from **bedtime movements**, the derived **sleep-wake (fragmentation) patterns**, as well as other extracted **time series-based features**. To achieve the aforementioned biomarker bundle, SInBAD first needs to undergo a **massive calibration phase** in which the **sleep episodes** of people in **very preliminary stages of cognitive impairment** will be **long-term monitored**. Over time, the prognosis of these people will evolve (with an estimated degree of probability [2]) towards an AD-grounded dementia (or not). Some of them will get better with time and the rest will experiment no relevant variations regarding their cognitive status. These specific **mental health conditions will be regularly assessed and monitored by a panel of mental health experts** and for each volunteer. The main tool will very like be cognitive assessment tests, which have proven adequate enough for this task [3].

In a next step, modern **machine learning** (ML) methods will be used to characterize the specific features (i.e., the sought biomarkers) that best discriminate each aftermath. These methods will in turn leverage the same techniques that have been formerly applied with success in time-series, causal inference, **atmospheric-ocean** state, and **weather prediction** [4]. **Hence the maritime nature of the acronym of the project: SInBAD**. These inertial biomarkers will be presented to the mental health research community as and adjacent, complementary, and **easy-deployable tool aimed at improving the early detection of Alzheimer's**. In this regard, this project also fits quite amply into the domain of **translational research**.

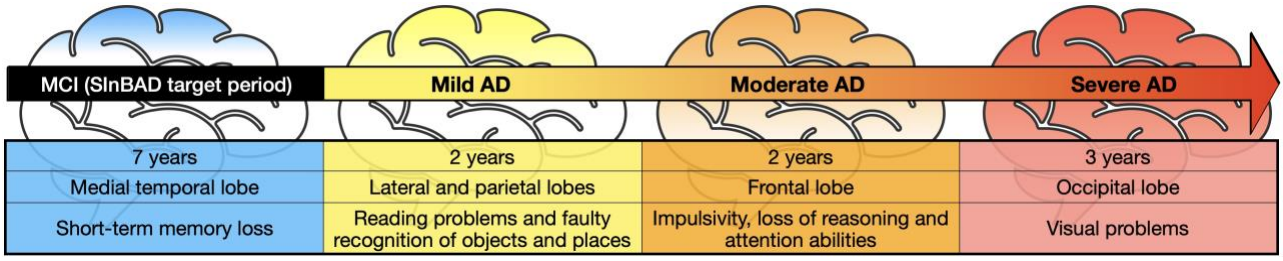

Figure 1: Evolution from MCI to Severe AD

Biomarkers and Alzheimer: still on training wheels?

Biomarkers are tools that allow us to identify the dawning or the specific stage of a given syndrome or disease. Some biomarkers are substances or methods very closely related (from a biological and even spatial point of view) to the health disorder being explored, and thus, they may require **annoying** (and even painful) **physical probing** of the patient's body. Since the identification of AD by Alois Alzheimer [5], many biomarkers have been tentatively devised and proposed by the neuroscientific community with the goal of staging the level of dementia, and when feasible, achieving an **early identification** in order to take preventing actions. Current AD biomarkers are some **invasive** in relation to habits, and privacy and even though some of them do not imply the minor discomfort for the average person, it might not be quite the

situation for someone starting to manifest MCI. As it has always been consistently acknowledged by family members, caregivers and therapists, **AD patients may experience episodes of aggressiveness** and explicitly externalised irritation, even at very **early stages of the disease** (prodromal phase).

Aggresive behaviours have also been identified from a psychiatric point of view [6]–[10]. As it might seem obvious, they often turn the most elementary diagnostic/evaluation tests into unpleasant experiences at hospitals, clinics, labs, and during ordinary medical consultations. **Even inoffensive cognitive assessment sessions can end up in adverse terms**. The situation worsens in the case of early-stage AD patients because patients may feel uncomfortable, terrified, and somehow defied by the necessary intellectual effort he/she is no longer able to deliver. In order words: physically probing or interviewing a person with this type of dementia can be **an experience that involves unnecessary inconvenience** (and even real risks) for the patient, accompanying relatives/caretakers, and for the health or scientific staff.

In addition to these operative limitations, AD is a disorder that begins its development years before the very first clinical and behavioural signs start to be recognisable. Sometimes, this growth is quick and appalling. On other occasions, it is a process that takes years to become noticeable. For this reason, it seems very convenient to define **biomarkers that follow and remain functional during this slow disease pace**.

The chicken-and-the-egg sleep/AD causality dilemma

Sleep science and Neurology have recently achieved remarkable contributions to the understanding of the link between AD and sleep [11], [12]. From an endocrinological perspective, the **lymphatic and glymphatic systems seem to act as a cleansing mechanism for the brain**. Most importantly, these purification pipelines and channels seem to work at full steam **only during sleep**. A graphical summary of this chicken-and-the-egg causal chain of events that may take part in the Alzheimer/sleep interplay is shown in Fig. 2.

These recent findings have made it very hard for the scientific community to escape the conclusion that **there is a tight connection between sleep** patterns and the dawning (and development) of an AD case. Although very laudable, the scope and ambition of all the aforementioned research efforts ceases at the sleep/AD binding determination and quantification. We respectfully think that **the bond between night-time conducts and Alzheimer's can be better leveraged** and that a new set of unobtrusive bedtime biomarkers can be distilled for this specific mental disorder. These new fiducial system can eventually **contribute to the assessment and identification of very early AD** scenarios and subsequently trigger available treatments or therapies aimed at diluting and/or delaying the onset of symptoms.

Besides, the **monitoring** of bedtime behaviour has been largely boosted in recent times, as well as **simplified**, from a technical and operational point of view. **Previous** sleep tracking methods relentlessly required complex and **intricated field equipment** (i.e., wired, composite and entangled polysomnographic hardware). In this *legacy* experimental context, and as also takes place with the aforementioned biomarker extraction techniques (CT, CMP, CSF, etc.), sleep patterns are just impossible to derive from a person suffering from (or beginning to sense the effects of) AD. In other words, the **state of alteration associated to this dementia make it very awkward to safely and appropriately monitor a person's sleep** with former sensors and devices. In the case of unattended research scenarios (e.g., the patient's home), in addition to very likely disturbing the person's sleep, the used devices can risk being damaged or simply lost. Furthermore, from a pure scientific point of view, we have to completely rule out the possibility of performing such surveys in a continuous and unsupervised manner (i.e., every night).

At this point, it is **impossible to escape the following line of reasoning**. First, science has demonstrated that AD and sleep are very intimately linked (**causality loop**). Second: sleep-tracking technology has dramatically evolved in the past years. Third: MCI can be a long phase in which bedtime patterns may begin to warn us about the mental status decline to AD (or not). From all these evidence, it is obvious to conclude that continuous bedtime tracking and analysis should be better leveraged and applied as an **ahead-of-time magnifying lens** to look into the patient's brain (i.e., a very powerful and down-to-earth AD biomarker).

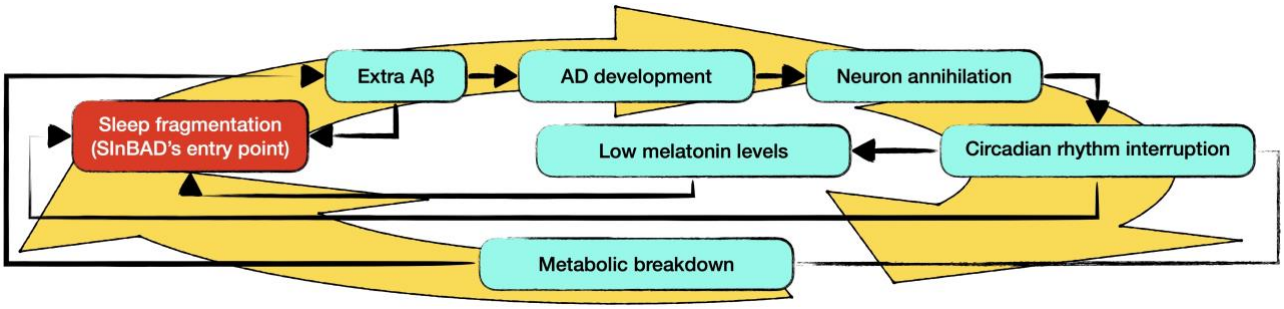

Figure 2: Theorized pathway (and positive feedback loop) between deficient sleep and Alzheimer's

The loophole: feasible long term sleep monitoring in MCI

As stated above, **bedtime observations are nowadays a very plausible** practice that can be quite easily **standardised** and regularised. There exist plenty of solutions and techniques (motion detection from video inputs, ultrasound echo, smart Wi-Fi mapping, tailored/disguised polysomnography systems, etc.), but they are still some incompatible for the tackled scenario and due to the reasons already presented (**privacy**, still some minor technical complexities/difficulties on the side of relatives/caretakers, etc.). An added adversity is the extremely **long length of some subtle studies**, which may need to be executed for months or years.

In the current proposal, we suggest the use of simple **sensors** that continuously log raw acceleration (**inertial**) information at a given frequency. These devices can be extremely **tiny**, clean and go **unnoticed** by the person wearing them. These equipment are also popularly known as *actigraphy* devices in the consumer market tier. In contrast with the versions targeted at the general user, the plan is to use simple **scientific-rage autonomous sensors that log low-level information about the nightly movements** of the wearer (and which are in linear relation to the **disruption of sleep**). These devices fit better into our *delicate* research scenario and may represent the beginning of the needed **loophole into the long sought effective AD staging**. They are also **extremely simple to use, even for senior people** with zero knowledge on digital devices or gadgets. The same applies for collaborating staff at nursing homes.

Besides, as widely demonstrated by many research works, it is possible to **derive sleep and waking periods from raw inertial data** [13] as well as many other relevant time series-related features from both types of cycles [14]. Finally, although we may seem to have reached a definitive methodology regarding the appraisal and logging of bedtime patterns, SInBAD is **open to explore alternatives and study more appropriate methods** to fulfil/complement this task. The proposed interplay of modern, easy-to-implement and **unobtrusive hardware** is the perfect tool for **sleep monitoring in adverse settings** . The simplicity of these systems makes them ideal for massive use in longitudinal and cross-sectional studies.

Challenges ahead

The core objective of the project is the progress towards a **deeper understanding of the sleep-related mechanisms that anticipate a cognitive decay from an MCI status**. The main challenges are:

1. Gather a complete (and electronically open) **map of the sleep** patterns for MCI.
2. Define a compromise, long-term **sleep monitoring standardised methodology** for MCI scenarios.
3. Thoroughly study, select and adapt the **most appropriate sleep-tracking devices** and methods.
4. **Integrate** data streams and metadata into an efficient and **standard storage** system.
5. Distill those **chronofiducials** that best foretell the **advent of yet unobserved cases of AD**.
6. Promote and **disseminate** the use of these new AD markers among the medical community. In other words, make of SInBAD a true **translational research effort**.

Materials and methods

This project integrates **sleep science**, **wearable** technology, **neurology** and **machine learning**. To limit the risk and guarantee the success of the project, the research plan has been designed with a highly iterative approach: the basic lessons learned after each iteration will be applied in the design and control of the process of the next cycle (Fig. 3). As with every frontier research in the realm health+machine learning, it is very difficult to know in advance the amount of necessary data to collect (measured as *person-months* units). For this reason, a **continuous data analysis approach** will be followed. This technique is also known as *Continuous Delivery for Machine Learning* or CD4ML [15]. In this context, sensor data will be downloaded, registered, tagged and inspected in a recurrent manner in order to **optimise the data collection phase**.

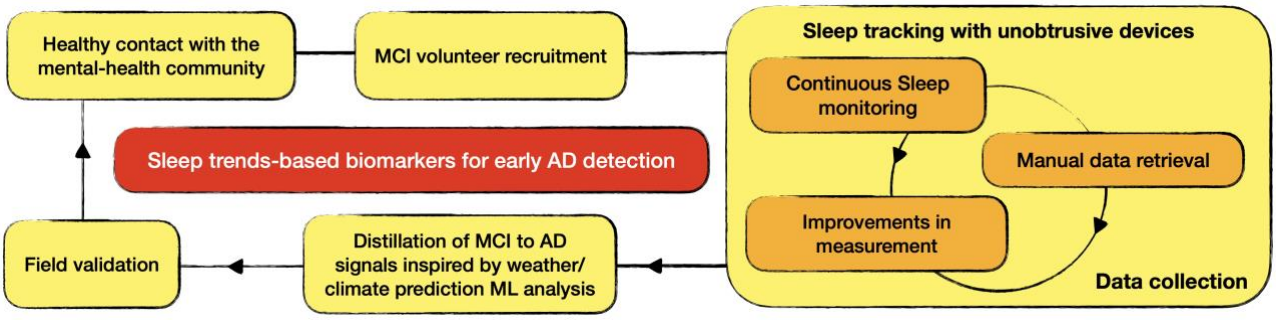

Figure 3: Project workflow that takes into account a CD4ML methodology

Data collection phase

Gathering sleep data is the first and most critical phase of our setup. It is therefore important to build a large committed cohort of volunteers. The composition of this group should also maximise heterogeneity regarding the possible mental health aftermaths of the participants. This phase also implies transversally **monitoring the sleep periods** of mild cognitive impaired seniors, regardless of their age, gender, cultural level, ethnicity and social background. Of course, these data will be **anonymised** and logged for the sake of completeness, as well. Besides, all participants will fill-in and sign and **informed consent** form whose format and contents will be agreed between the host institution and external collaborators. In this sense, **the project will comply with all current regulations regarding data protection and bioethics**.

Volunteers will be recruited with the kind help and agreed participation of neurologists, medical centres, mental stimulation institutions and retirement homes. In the case of non-inpatients, typically, a person starts experimenting a *subjective cognitive impairment* (or SCI). Next, he/she is examined by one of the aforementioned specialists who issues a diagnosis. According to previous studies [16], approximately half of the people who begin complaining about SCI end up declining to MCI in the following years. The expected time of monitoring will be a non-negligible fraction of the agreed average time span for the MCI stage (prodromal AD). This period does in fact vary from case to case, but a mean value can be established around 2 years after the preclinical phase has ended. For this reason, a **6 continuous months compromise range** has been tentatively determined, although this parameter will be revised as the project evolves following the CD4ML philosophy. During this period, the sleep behaviour will be monitored every night. In this context, the **intervention and help from relatives, nursing institutions and professional caretakers is crucial**. In a very similar way, the **number of participants will be iteratively revised**, although an initial number of 1000 participants has been considered and has been accordingly taken into account in the proposal's budget and timeline.

Long term sleep tracking with unobtrusive devices

Independently of the chosen sleep-tracking method/device (which represents one of the challenges of this project), the fundamental requirements for long lasting sleep monitoring scenarios are: **unobtrusiveness**, large operation **lifetime** and **unwired** design (no connection to the Internet). This last property enforces the **privacy rights** of the participants (including that of relatives and accompanying healthcare professionals, as well), which represents a very important aspect in SInBAD. **Ease of use** (by senior people or just anyone with aversion to technology) should also be considered. A good preliminary candidate that fulfils these requirements is the AX3 **accelerometer** from Axivity, Ltd. Notwithstanding, other available devices will be explored over time, and when its use seems pertinent, they might be incorporated to the experimental setup as long as they can be integrated into the system from a technical and physical point of view.

In addition to the objective actigraphy streams, **clinical** and social **data** (health status, past or present diseases, medication, surgical operations, etc.) will also be, after an **anonymization** process, accordingly and periodically collected. A special attention will be paid to sleeping medications. All recorded data will be stored and made **openly available to the the AD and sleep research communities**. In this step, **electronic standards for medical information storage and transmission will be used**. In SInBAD, it is important to follow and respect internationally agreed protocols and procedures for the digital preservation, securitisation and sharing of health records. Sadly, this is not the common case for the domain of life sciences research, where these praiseful agreements (DICOM, HL7, EDF, etc.) among industry leaders are often ignored.

Biomarker distillation: forecasting the AD storm

Accelerometers (as well as other registered sensors) log raw **physiological data streams**. Depending on the sampling frequency, we can be talking about several Tbytes/person. At the end, the acquired features are **very similar to those measured by weather stations**: wind speed, temperature, humidity, etc. Both experimental scenarios can also compile specific discrete quantities (e.g., number of awakenings in a given night vs. number of outlying gusts of wind in a given day). Formerly, **weather prediction** relied on classical statistics-based methods [17]. Since recent times, meteorology has embraced machine learning (ML) [18]. The interesting point here is that **the same modern ML methods can be applied to both realms**: sleep/human activity recognition and weather forecasting. These shared ML techniques mainly comprise convolutional and long short-term memory recurrent neural networks and their goal is the same: to report upcoming weather changes, ahead-of-time. Likewise, given that a looped causal relationship between AD and sleep has been certified by neurology, **structural causal models** [19] will be applied as well.

In summary, in this phase we will distill the sought set of chronofiducials for early AD diagnosis. This computational process will be heavily inspired by the same state-of-the-art methods that are currently being successfully applied in the domain of weather, climate change forecasting and **causal inference**.

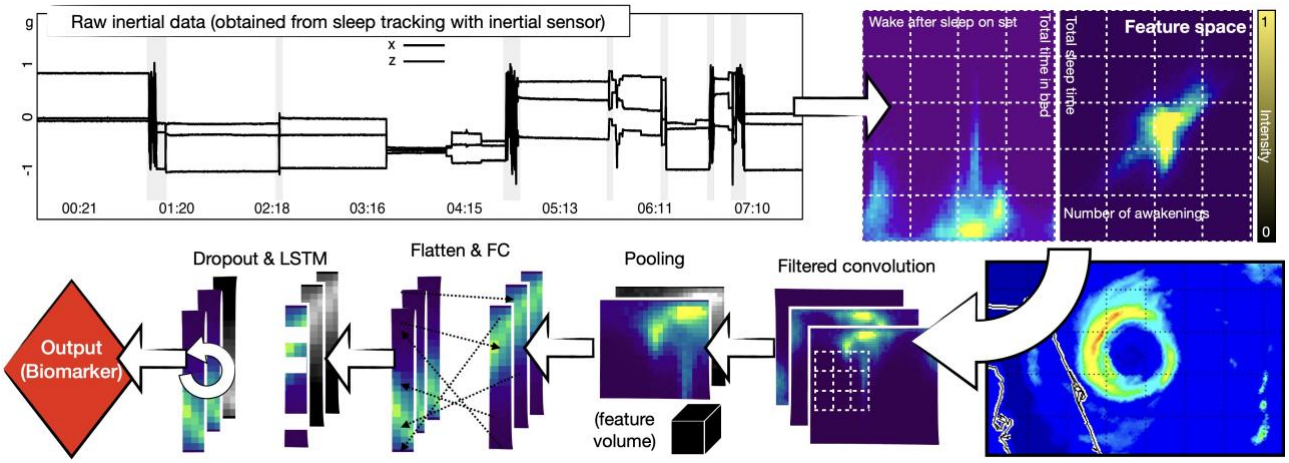

Figure 4: Representation of a neural network for sleep biomarker distillation (and local weather forecasting!)

Validation

Part of the aforementioned CD4ML strategy will also enforce the validation of the distilled biomarkers. For this reason, in a second phase of the project, **the determined set biomarkers will be tested** with an external cluster of volunteers. This task will be carried out in close collaboration with the same of group neurologists, psychiatrist and psychologists that will also assist us in the first stages of the project.

Dissemination and outreach

Sadly, many successful research do not end up permeating society. This occurs when lab or theoretical efforts are not properly transferred to the real world (professional associations, guilds, authorities, etc.). In fact, the study of AD is a very vibrating research area, with many publications and daily advances in a myriad of topics that go from prevention techniques to treatments and therapies. However, many of them are never translated into **pharmaceutical products**, nor do they end up being **part of therapeutic protocols**.

For the aforementioned reasons, **an important part of SInBAD is the dissemination of its results among the scientific community**. Nevertheless, these outreach tasks will not stop there. Psychologists, GPs, family therapists, as well as other professional associations will be informed of the new proposed sleep-grounded biomarkers. These **dissemination efforts will go beyond conventional outreach events** and media (scientific congresses, journal, workshops, etc.). The SInBAD project will carry out specific actions that emphasise the application of the proposed complementary early AD detection mechanisms among psychiatric consultations, health clinics, sleep disorder units and health professionals in general, etc.

Opportunities and breakthroughs

So far, what seems to be confirmed is that there is give-and-take toxic alliance between AD and sleep and that **Alzheimer's patients sleep differently depending on the stage of progression of the disease**. SInBAD goes beyond this solid evidence and proposes the incorporation of new sleep- and inertial-based biomarkers for early AD detection. The required measurements on a person being assessed are performed by night. In addition, the probing devices are extremely transparent to the person being diagnosed. **The possibilities offered by SInBAD were inexistent until a few years ago** for two main reasons. On one hand, the landscape of motion sensors had to significantly evolve in order to be felt completely unobtrusive by the target wearers. On the other hand, computing power and machine learning techniques had to still undergo a democratisation process that would enable the distillation of large amounts of high precision motion data. These two challenges can be summarised into the following one: **sleep science**, in general (and polysomnography equipment in particular) still **needed a miniaturisation revolution**. Once these challenges have been overcome, it is time to apply these new possibilities to ageing science realm.

In this context, SInBAD wants to be the **spearhead in the area of AD prevention through a practical sleep tracking approach**. Neurology in general should join this new exciting wave linked to *sleep science for the rest of us*. In contrast with the approach followed by consumer devices and solutions, Alzheimer's is a very serious disease that should be studied, diagnosed and treated from a clinical perspective. SInBAD will embrace this perspective by providing a realistic framework that can still grow through **future collaborations** and with the common goal of improving the quality of life of millions of people through the **benefits of early diagnosis**. SInBAD is also opening the door for future research and advancing the use of **ubiquitous technologies** with a different and innovative perspective. With time, these efforts (and associated results) could be the base of future **patents**, **products** and health-oriented **services**.

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Section b: Curriculum vitae

PERSONAL INFORMATION

Name: Corbi Bellot, Alberto

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URL: https://en.unir.net/profesores/alberto-corbi-bellot/

Nationality: Spanish

EDUCATION

2017 PhD in Medical/Particle Physics (mark: *excellent Cum Laude* + several awards). University of Valencia (UV) + Institute for Corpuscular Physics.

2009 Master of Advanced Studies in Atmospheric Physics (mark: *excellent Cum Laude*). Catholic University of Valencia (UCV)

2003 Graduate in Physics (5-year degree equivalent to Grade + Master)

CURRENT POSITIONS

2013 – Assistant professor/lecturer at School of Engineering of Universidad Internacional de La Rioja (UNIR)

2020 – Director of the Degree in Physics at UNIR

2013 – Senior researcher at the Research Institute for Innovation & Technology in Education (UNIR iTED)

PREVIOUS POSITIONS

2003 – 2004 Researcher and teacher at the Institut für Kernphysik (Johannes Gutenberg-Universität, Mainz), A4 Group, MAMI synchrotron

2006 – 2008 Cofounder and developer at Kanteron Systems, Ltd.

2008 – 2009 Manager of Environmental Forecasting Systems for the Spanish Ministry of Infrastructures and Public Works (through the outsourced company SIDMAR)

2009 – 2013 Cofounder and developer at Kanteron Systems, Ltd. During this period, I was involved as PI in several R&D projects funded by public entities: AddBio (CDTI), PathoPACS (CDTI), DiagnoPOC (IMPIVA)

2013 – 2017 Researcher at Institute for Corpuscular Physics (IFIC), part of the National Research Council (CSIC)

FELLOWSHIPS AND AWARDS

2017 – 2019 European Radiological Society (ESR)

2017 – Alzheimer Family Members Association (AFA)

2015 🏆 Best dissemination project (awarded by the National Center for Particle, Astroparticle and Nuclear Physics)

2015 🏆 University-Society AlumniPlus (awarded by the Governing Council of the University of Valencia and the Valencian government)

2017 🏆 Best research paper (editor's choice) by the journal Medical Physics https://doi.org/10.1002/mp.12144

2018 – Member of the Spanish Royal Physics Society (RSEF)

2010 – Member of the International Aikido Federation (AIKIKAN)

SUPERVISION OF GRADUATE STUDENTS AND POSTDOCTORAL FELLOWS

2013 – Supervision of over 60 Graduate and Master thesis works at UNIR

2016 Supervision of a graduate student at UPV (Polytechnic University of Valencia)

2020 Supervision of a graduate student at UNED (National Distance Education University)

2019 – Currently supervising 3 PhD students at UNIR

TEACHING ACTIVITIES

2013 – "Physics for computing engineers" teacher at the School of Engineering of UNIR
2017 – 2018 Teacher at the Master's Degree in Software Engineering at UNIR
2013 – 2017 Teacher at the Master's Degree in Mobile Development at UNIR
2002 – 2003 Teacher at the Physics faculty of the Johannes Gutenberg University

ORGANISATION OF SCIENTIFIC MEETINGS

2007 Member of the organisation: International Symposium on Marine Sciences (ISMS07)
2014 Member of the organisation: International Symposium on Computers in Education (SIIE)
2016, 2017, 2018 Member of the organisation: Congreso Internacional de Tecnologías Emergentes y Sociedad (CITES)

INSTITUTIONAL RESPONSIBILITIES

2013 – Director of the area of Master Thesis at the Master for Big Data Analytics and Unir
2017 – 2021 Global coordinator of Master Thesis area (Master's Degree in ML+ Big Data Analysis)
2013 – Involved in teaching innovation projects at UNIR (SAKAIING, HoTEL, uR-Tutor, AppMOOC)

REVIEWING ACTIVITIES

I currently perform reviewing tasks for the following journals and/or editorials:

2013 – IEEE Access (Institute of Electrical and Electronics Engineers)
2016 – ASOC (Applied Soft Computing)
2016 – MDPI (Multidisciplinary Digital Publishing Institute)

MEMBERSHIPS OF SCIENTIFIC SOCIETIES

2017 – Member of CEAFA (Spanish Confederation of Relatives of People with Alzheimer's)
2017 – 2019 Member ESR (European Society of Radiology)
2005 – 2013 OsiriX Foundation (DICOM Medical Imaging Software and part of the NEMA)

MAJOR COLLABORATIONS

2016 – Writer for Muy Interesante (monthly popular science magazine, 350k copies worldwide). I have published several dissemination articles
2011 Speaker for the Colombian Radiology Association
2012 Project collaborator at University of Las Palmas de Gran Canaria (Dept. Signal Processing)
2013 – I have been involved in many outreach activities in radio stations (Capital Radio, Cadena SER), television (Spanish Radio and Television Corporation, RTVE), and press (UNIR Revista, Noticias de La Rioja, The Conversation)
2018 Speaker at Universidad Nacional Abierta y a Distancia (UNAD), Colombia
2016 Speaker at GMEPE (Global Medical Engineering and Physics Exchanges)
2017 – Scientific collaboration with UNED (National Distance Education University) and involvement in publicly funded projects (project INT2AFF)
2018 – Scientific collaboration with University of Salamanca (public funded project DDOL)
2015 – 2018 Research collaborator at ENRESA (Empresa Nacional de Residuos Radiactivos)

CAREER BREAKS

2004 – 2006 Serious illness (cancer) and eventual death of family member
2017 – 2018 Serious illness (Alzheimer) of family member

Section c: Early achievements track-record

With a background in physics, my passions have always been scientific **outreach**, **environmental** and **medical physics**. In fact, my last scientific effort has been the SQUAD project (**Sleep Quality in Alzheimer's Disease**). SQUAD has demonstrated, among other things, to be a **firm proof on concept on the possibility of objectively tracking sleep patterns in people in all possible stages of AD**. I humbly have to admit that this research work has evolved in an extraordinary positive way and that **has exceeded my most optimistic expectations**. As a matter of fact, this is the main rationale behind the present ERC StG proposal: **evolving this fresh AD⇆sleep line of research to a new level of maturity and solid clinical/social usefulness**. My former laboral, R&D, business and academic trajectories (summarised next), have turned-out to be the **cornerstones behind my current proficiency regarding the fields of sensors, health, environmental physics and ML** and have allowed me to devise the SInBAD project.

Track 1 (predoctoral): Particle & Environmental Physics and Medical Imaging

After graduating in Physics, I was offered a two-semester research and associate professor positions at the Johannes Gutenberg University. There, I worked in one of the **experiments that were run by the Nuclear Physics Institute (KPH)** around the 1.6 GeV MAMI microtron. This opportunity allowed me to gain knowledge about electronics, data processing, lab management and consolidate attitudes towards experimental correctness, scientific rigor and teamwork. My stay also granted me a vast accessory **awareness around radiological protection and medical physics**.

A few years later (after career break), **I cofounded Kanteron Systems**, a company focused on **R&D projects in medical imaging, hospital informatics and healthcare technology**. Kanteron is operative to date, and although I no longer hold an admin role, I still have an informal and friendly relationship with the board (oriented at technological counselling). Needless to say, this business-focused period enabled in me the acquisition of important competencies which have later turned-out capital in healthcare-related projects: **sensitivity towards patients staff** and medical institutions, pursuit of excellence, etc.

During this period an as part of my MSc of Advanced Studies (Spain "DEA"), I also worked for the **Ministry of Public Works** managing the infrastructure of **environmental sensors** (buoys and tide rangers). This stage of my academic life, although somewhat offbeat from the medical/particle physics field, allowed me to gain supplemental insight towards time series data, sensors, **forecasting**, and the **coordination of field service work**. This gained know-how has unquestionably inspired the current proposal.

Track 2 (doctoral student): Medical Physics and Educational Technology

While still at the business realm, I decided to start my **PhD in Medical Physics at the Institute for Corpuscular Physics**, part of the National Research Council. My research topic, although grounded on particle detectors combined with environment sensors, was also very focused at the engineering and product-delivery levels. My PhD also permitted me to continue having a tight and **daily contact with health care institutions, patients and medical settings**, which was starting to become vital for me. In addition, I was lucky that my two PhD supervisors came from different disciplines: particle physics and image processing/telecommunications, respectively. From the academic and scientific production point of view, this was a very successful period: **5 papers and 2 PCT patents** [1, 2, 3]. I was also granted 2 prestigious **academic awards** based on my research. All these achievements share a common scientific denominator: the tangible **interplay of contour and motion sensors with particle/medical physics settings**.

Meanwhile and appealed by the academic life, I combined the aforementioned duties with a **professorship at Universidad Internacional de La Rioja (UNIR)** and slowly became interested in field of ICT (with a special focus on Physics). For this reason, **I joined the UNIR iTED group as a senior researcher**. This group and academic institution (UNIR), although still very young, have always held the doors open for me to embrace new personal projects. Certainly, some of the research activities carried out within UNIR iTED have been focused on the field STEM education. Nevertheless, since the start of our collaboration, I have also been **granted a high level of confidence and independence when proposing new research topics as long as they would lead to social awareness** on a specific or given subject.

[1] F. Albiol, A. Corbi, C. Bellot and A. Albiol, "Device for Extracting Three-dimensional Information from X-ray Images of an Object, Method for Calibrating said Device, and Method for Generating said X-ray Images," Patent WO2016156646A1, (2016) http://bit.ly/corbirxpct

[2] F. Albiol, A. Corbi, and A. Albiol. "Geometrical calibration of X-ray imaging with RGB cameras for 3D reconstruction," *IEEE transactions on medical imaging* 35, no. 8, pp. 1952-1961, (2016).

[3] F. Albiol, A. Albiol, A. Corbi, J. Agramunt, L. Caballero and C. Domingo, "System and method for the volumetric and isotopic identification of radiation distribution in radioactive surroundings," Patent WO2018146358A1, (2018) http://bit.ly/corbigammapct

Track 3 (postdoctoral): ML, Human Activity Recognition and Sports Physics

In this context and driven by my devotion towards sensors, motion tracking, physics and health, I **joined forces with the aDeNu research group** (Adaptive Dynamic online Educational systems based oN User modelling) from UNED Uni. This fruitful collaboration resulted in a very interesting research line on **sport physics and user modelling** that still continues to this day: several papers and workshop presentations have been presented, and there are still plenty of outcomes in the pipeline yet to be introduced, as well as several MSc theses. Within this period, I also assumed the role of coordinator of MSc thesis works at the Master in Machine Learning+Big Data at UNIR. Although not linearly linked to the current ERC project proposal, **the techniques learnt and applied within these ML-related areas were silently paving the way for the operational scheme followed in the SQUAD initiative** (tackled next), and for the present StG outline.

[4] O. C. Santos and A. Corbi, "Can Aikido Help with the Comprehension of Physics? A First Step Towards the Design of Intelligent Psychomotor Systems for STEAM Kinesthetic Learning Scenarios," in *IEEE Access*, vol. 7, pp. 176458-176469, (2019).

Track 4: Contributions to Alzheimer's Research

My second career break inadvertently immersed me in the world of Alzheimer. After performing the appropriate literature research, it swiftly dawned on me that there was very **little work done around clinically observed sleep patterns in AD patients and the destruction forged by this syndrome**. Certainly, a narrow and **causal connection to circadian rhythms** and nightly biological processes seems to have been thoroughly stablished in recent years. Nevertheless, I found peculiar that almost no transversal, massive and objective bedtime studies had been carried out to identify behavioural biomarkers based on the continuous and preventive monitoring of sleep patterns. I felt I was in front of huge a **research niche that deserved been explored** with a new perspective and that I could bring all my formerly acquired knowhow (medical physics, sensors, weather/time series prediction, patient care, etc.) to the game.

As a member of the Spanish Alzheimer's Confederation, in 2018, I decided to join forces with the some reputed and **retirement homes and mental stimulation centres**. Together with the support of UNIR iTED, we devised a tailored and proof-of-concept research scenario that would later pave the way for a more long-winded scientific project. This is how **SQUAD** started. The project's goal was to **re-identify this recent AD⇆sleep marriage but from a pure nocturnal motion-analysis and clinical perspective**. That is, instead of assessing the sleep status and mental condition with complex equipment and the acquisition of painful physical evidence, **SQUAD makes use of a simple operative design**. On one hand, sleep cycles are tracked with **unobtrusive motion capture** hardware: an AX3 accelerometer inside a comfortable wrist-band enclosure. On the other, the cognitive deterioration condition of each participant is gauged with a more traditional tool: the ordinary **clinical follow-up by psychologists**. Needless to say, this experimental compromise setting has its limitations and its degree of uncertainty. However, after successfully measuring several nights of **sleep of over 180 people in different stages of AD**, plus 40 from healthy people (**more than 800 nights**), we can conclude that the devised empirical setup has turn out a very good research compromise. The preliminary results have been published in [5]. In this research paper, we applied a modern **ML analysis** to the raw inertial collected data during sleep and determined it was possible to satisfactorily **cluster >85% of the participants (women) according to the severity of their cognitive status**. As also exposed, this agreement seemed to be quite independent of age and other health conditions. There is **another paper being prepared** for submission to an IEEE journal in which we show how the very same technique can be profitably leveraged to **filter the two most extreme cognitive degradation stages** (leves 6/7, GDS scale). An outreach article in the monthly popular science magazine Muy Interesante was also published.

As stated before, UNIR has generously blessed me over the years with an increasing amount of trust and independency towards **my research career**. With the support UNIR iTED, I have been able to **blend my former expertise in physics, environment and health with social awareness**. An StG will allow me to **delineate a new research group fully devoted to the field of sleep research in AD**. I firmly believe that **there should exist new biomarkers already present in the bedtime behaviour**. These *chronofiducials* should be somehow buried in the inertial time series stream, waiting to be distilled through tenacious data compilation+ML analysis procedures. I am also convinced of their potential to point at a dawning of AD cases months or years before the first signs of dementia and earlier than any other diagnosis method.

[5] A. Corbi and D. Burgos, "Connection between sleeping patterns and cognitive deterioration in women with Alzheimer's disease," in *Sleep and Breathing*, Springer Nature, 2020.